DSC 207 / PROJECT MILESTONE

Preparing gravitational-wave data for inference across observing runs

Ali Lordifar
ali_lordifar@berkeley.edu

Kennedy Goliman
kennedyg@berkeley.edu

Noah Fonck
noahfonck@berkeley.edu

Aaron Rassiq
aaronrassiq@berkeley.edu

Motivation and checkpoint scope

Our project asks whether a model trained on O3a LIGO noise can infer binary-black-hole properties reliably on O4a noise, including trustworthy uncertainty. Detector noise and the population of detectable signals can differ across runs, so preparing comparable inputs is essential. This milestone focuses on the dataset, preprocessing, and exploratory analysis. This report defers model comparison and uncertainty findings to the next stage.

Data prepared so far

The existing pipeline contains public GWOSC Hanford (H1) strain from eight quality-checked, 1,024-second segments per run at 4,096 Hz. Together, these provide **16,384 seconds and 67,108,864 raw strain samples**. IMRPhenomD waveforms are added to real backgrounds to obtain known labels. O3a required 1,375 attempts and O4a 847; SNR filtering rejected 575 and 47, leaving **800 accepted injections per run**. No waveform-generation failures were logged. Time samples are not independent labeled examples.

Saved allocation	Examples	Final input
O3a development pool	750	Four seconds of processed strain: 16,384 values
O3a held-out evaluation	50	Same input representation
O4a cross-run evaluation	800	Same representation; excluded from fitting

Each example targets chirp mass (solar masses), mass ratio, effective aligned spin, and distance (Mpc). These are saved dataset allocations, not a new manual rebuild. The previous estimator split its 750-row pool internally; exact fitting/validation counts were not recorded. For the next comparison, we plan an explicit 600/150 fitting/validation allocation within that pool, retaining the 50-row O3a holdout. This split remains planned.

Preprocessing applied to the saved dataset

Reject invalid/nonfinite segments rather than impute missing strain. Use data-quality checks, a one-hour exclusion around cataloged events, and a broadband standard-deviation screen (10^{-19} to 10^{-17}). Estimate a PSD from each original noise segment with four-second Welch windows. Whiten the padded injected buffer with that background PSD, apply a fourth-order 35-350 Hz zero-phase bandpass, then crop two seconds before and after the merger. No categorical encoding or learned feature selection is applied. Whitening is the signal normalization step; any model-specific scaling will be fitted on training data.

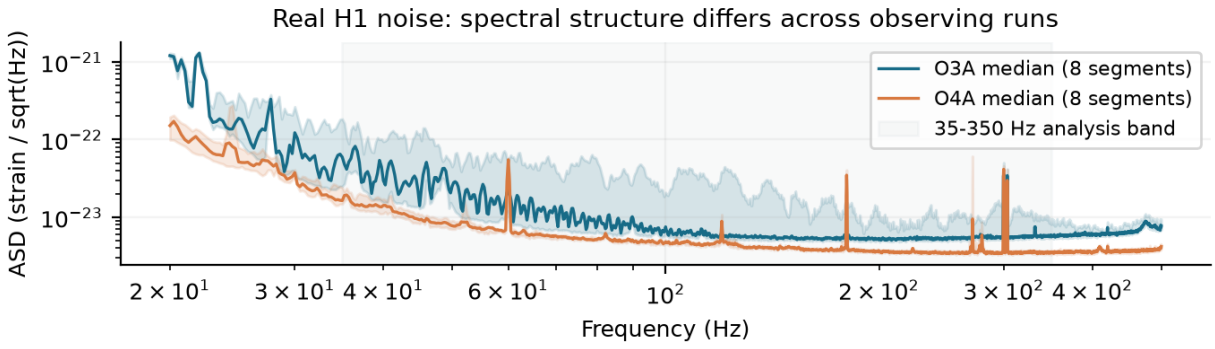


Figure 1. *Median ASD and interquartile shading across eight segments per run show frequency-dependent structure and within-run variation. O4a has lower noise amplitude over much of the displayed band. These differences motivate consistent preprocessing and later cross-run evaluation; the curves alone do not establish model degradation.*

Exploratory analysis and data quality

These checkpoint-based figures summarize the prepared population and issues to address before model comparison.

Target distributions after selection

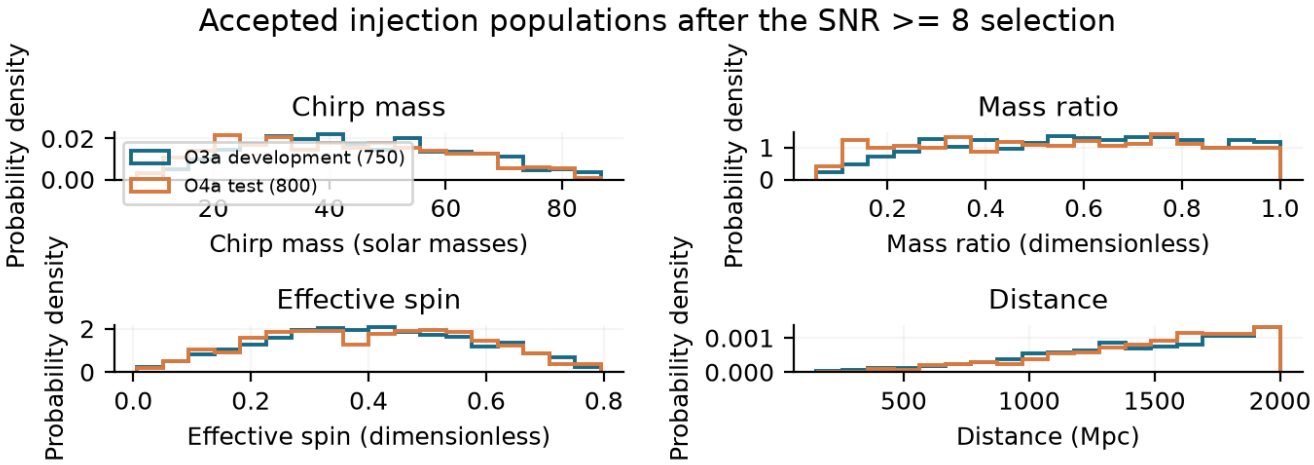


Figure 2. *Distance is concentrated toward larger values, consistent with the implemented Euclidean volume-weighted draws. Chirp mass and mass ratio are nonuniform after deriving them from ordered component masses. Spin is positive because component spins lie in [0, 0.8]. The same draw rules yield different accepted populations: O3a and O4a SNR rejection rates are 41.8% and 5.5%.*

Relationships between inputs and targets

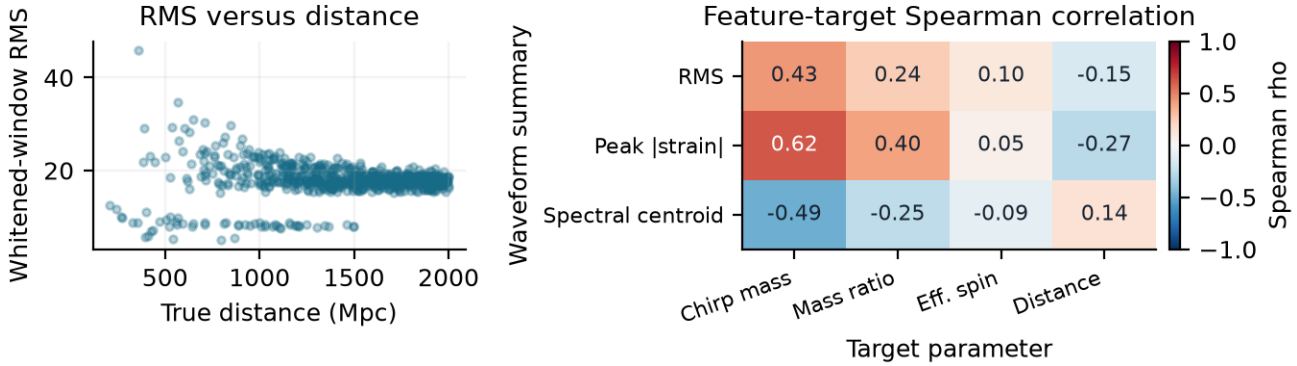


Figure 3. *Each point is an O3a development example. RMS alone weakly distinguishes distance (Spearman rho = -0.15). Peak absolute strain is more strongly associated with chirp mass (rho = 0.62), while spectral centroid is negatively associated (rho = -0.49). These summaries identify potentially useful relationships; correlation is not model feature importance.*

What the EDA tells us

The saved target correlation analysis finds chirp mass versus mass ratio r = 0.61, consistent with their shared dependence on component masses. Time-domain exploration includes full strain, merger-centered zooms, and before/after whitening; no systematic within-run time trend has been established. With 16,384 inputs and 750 development examples, reducing variance will matter. Model-based feature importance has not yet been measured; validation-set permutation importance on compact waveform summaries is planned and remains a rubric item to complete.

Challenges and current treatment

Shared noise: many examples reuse eight backgrounds per run, so the existing example-level O3a split does not demonstrate independence of source noise. A grouped-by-source sensitivity check is planned. **Selection effects:** compare parameter/SNR distributions before attributing differences to noise alone. **Valid loud observations:** retain physically plausible extremes rather than clipping them indiscriminately; distinguish these from invalid data. **Boundary effects:** process padded buffers before fixed-length cropping. **Known candidate time:** merger-centered inputs assume a supplied event time; this is parameter estimation, not blind detection.

Next experiments and milestone handoff

Methodology: keep the main modeling direction simple

Linear regression provides a point-estimate baseline, and SNPE with a masked autoregressive flow provides the main posterior-estimation approach. To meet the rubric's three-algorithm requirement, we include Ridge as a small regularized extension of the baseline. The table describes the next common comparison; it does not claim that all three have been run on a newly prepared manual dataset.

Algorithm	Why it is appropriate	Next experiment
Linear regression - baseline	A transparent reference for mapping the processed waveform to four targets.	Fit on the explicit training allocation; compare with a training-mean reference.
Ridge - improvement 1	Adds regularization to the same linear approach, limiting overfitting without a new modeling direction.	Tune alpha over 0.1, 1, 10, 100 and 1,000 on O3a validation.
SNPE + MAF - improvement 2	Estimates a joint posterior, allowing uncertainty and parameter dependence to be examined.	Begin with a small controlled run, explicit validation counts and recorded training settings.

Validation, tuning, and evaluation plan

Use the same saved examples and target order for each model. Fit feature scaling on training data only. Select Ridge regularization and any representation choices on O3a validation; reserve O3a held-out data for within-run evaluation. O4a was already inspected in earlier project experiments, so further comparisons on the same examples are exploratory; fresh backgrounds are needed for an untouched final test. Audit source-group overlap and report actual counts after any revised split.

Compare per-target MAE, RMSE and R-squared, with units. Use a documented normalized error to compare across targets rather than letting distance dominate an aggregate raw-unit score. Choose models on validation performance, then report held-out results without retuning. For SNPE, add empirical 50%, 80% and 90% interval coverage, interval widths, and sampling-failure rates. Record seeds, hyperparameters, example counts and software versions. A small pilot is the next step; a completed full posterior study is not claimed at this checkpoint.

Prior and selection checks before calibration claims

The simulator draws component masses/spins and Euclidean volume-weighted distance; its derived four-target distribution is not a uniform rectangular prior. The SNR gate changes it further. Before interpreting posterior rank tests as formal simulation-based calibration, align the inference prior, simulator and selection assumptions. Use comparable posterior-sampling budgets across runs and report failed cases separately. This is a planned methodological check, not a calibration result.

What is complete, and what remains

Available now: saved real-noise segments, accepted injections, processed arrays and their provenance; raw time-domain inspection; target distributions and correlation analysis; the checkpoint-based figures in this report. The learning notebook also contains hands-on exploration of strain, waveform amplitude versus distance, and window preparation. The four role notebooks are task scaffolds, not evidence of completed individual contributions.

Next: complete the manually owned handoffs, freeze and record exact fitting/validation allocations, verify group separation, assess feature importance, and run the common baseline/Ridge comparison before extending the posterior experiment. Earlier model scores and calibration outcomes are intentionally outside this report's dataset/preprocessing focus.

Contributions and repository

Individual contributions and notebook owners remain to be confirmed by the team. Before submission, record each member's actual tasks and individually maintained notebook; each notebook must have one contributor and cell numbering starting at 1. This reserved section is included in the three-page limit.

GitHub: github.com/alilordifar/ligo-parameter-inference
Data: GWOSC public strain releases. Evidence: local checkpoint metadata, the September 12, 2026 UTC pipeline log, and saved notebook outputs. Counts describe those artifacts; planned allocations and experiments are identified explicitly.